

Research Article

Hierarchical Feature Recalibration Network for Motor Imagery Electroencephalogram (EEG) Classification

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Time–frequency–spatial (TFS) features play a crucial role in motor imagery electroencephalogram (EEG) classification. However, effectively leveraging these multidimensional features to enhance classification accuracy remains a significant challenge. Although feature selection techniques are widely used to extract informative TFS representations, most existing methods treat each dimension independently, overlooking the intrinsic grouping and hierarchical relationships among them. To address this limitation, this paper proposes a hierarchical feature recalibration network (HFRNet) that explicitly models intergroup dependencies and the hierarchical structure of TFS features, thereby substantially improving motor imagery EEG classification performance. HFRNet employs a two-layer weighting mechanism for hierarchical feature recalibration, followed by a classification module. In the first weighting layer, spatial features within each time–frequency segment are grouped and represented as feature maps. Channel-wise dependencies are captured through squeeze-and-excitation operations to learn channel weights, which are then used to rescale each feature map. In the second weighting layer, the recalibrated features are reorganized across time windows and further refined through a similar recalibration process. Finally, in the classification block, the refined features are flattened, concatenated into a single feature vector, and passed through dropout and fully connected (FC) layers for classification. Extensive experiments conducted on five motor imagery datasets demonstrate that the proposed HFRNet achieves the best overall performance, with an average accuracy (F1 score) of 81.03% (0.7931). Comparative evaluations against 30 feature selection methods and recent state-of-the-art approaches further confirm the superior effectiveness and robustness of the proposed model.

Keywords: EEG classification; feature recalibration; feature selection; motor imagery

1. Introduction

Extensive research has demonstrated a strong relationship between motor imagery electroencephalogram (EEG) signals and their temporal, spectral, and spatial characteristics [1, 2]. Consequently, time–frequency–spatial (TFS) feature extraction methods have been widely adopted for motor imagery EEG

classification [3]. However, effectively exploiting the rich information contained in TFS features to further improve classification performance remains a major challenge [4].

The primary technique for TFS feature learning is feature selection, which focuses on removing redundant information and identifying subject-specific TFS features. Filtered feature selection methods evaluate the importance of each feature

dimension independently, ranking them based on specific criteria such as the Fisher score [5], correlation coefficient [6], Kullback–Leibler divergence [7], mutual information [8], and minimal-redundancy-maximal-relevance [9].

Wrapped feature selection methods employ various search strategies to generate feature subsets and select the optimal subset based on evaluation metrics like classification accuracy and F1 score. Common wrapped methods include the firefly algorithm [10], genetic algorithm (GA) [11], binary differential evolution (BDE) algorithm [12], and binary particle swarm optimization (BPSO) algorithm [13], among others. However, these methods still treat each feature dimension independently during the search process. For instance, in the sequential floating forward selection method, features are added to the existing set one at a time [14].

Embedded feature selection methods differ from filtered and wrapped techniques by incorporating structural relationships between features. For example, the least absolute shrinkage and selection operator (LASSO) considers the sparse structure of features [15]. Variants like group LASSO (gLASSO) address group sparsity [16], sparse gLASSO (sgLASSO) captures both intergroup and intragroup sparsity [17], and fused LASSO accounts for continuous smooth structures [18]. These embedded methods generally outperform filtered and wrapped approaches in classification tasks because they take feature structural information into account [19]. However, existing embedded methods fail to address the hierarchical structure inherent in TFS features.

During TFS feature extraction, the raw EEG signal is divided into multiple time windows and then decomposed into several frequency subbands. Spatial features are subsequently extracted from each time–frequency unit [20, 21], forming a hierarchical structure of TFS features. In this hierarchical structure, each time window encompasses multiple frequency subbands, and each time–frequency unit comprises a set of spatial features. Notably, some informative time windows may contain less relevant frequency components, while key frequency bands may occur within less significant time windows. Hence, assigning hierarchical weights to both time windows and time–frequency units is crucial. Leveraging this hierarchical prior information can further enhance the classification performance in motor imagery EEG tasks.

Conventional feature selection methods often fail to capture hierarchical structures, whereas deep learning–based attention mechanisms offer a natural advantage in this aspect. Particularly, in domains like natural language processing [22, 23], hierarchical attention networks have proven effective for tasks such as document classification, exploiting hierarchical structures between words, sentences, paragraphs, and documents [24]. However, most existing hierarchical attention models are computationally intensive and require large datasets, limiting their applicability to EEG classification tasks, which typically involve small sample sizes and resource-constrained settings. To address these challenges, Hu et al. [25] introduced squeeze-and-excitation networks (SENet), which explicitly model channel interdependencies within feature maps for recalibration. SENet offers remarkable

performance with low computational overhead, making it easily integrated into existing network architectures to enhance performance. However, SENet is limited to recalibrating features within the same layer and cannot directly model the hierarchical structures inherent in TFS features.

It is noteworthy that many recent deep learning models enable end-to-end decoding of motor imagery EEG signals by integrating both TFS feature extraction and selection within a unified framework [26]. Representative examples include EEGNet [27] and Shallow ConvNet [28]. However, these models generally lack the incorporation of a priori information and often rely on complex architectures and large-scale training datasets. In contrast, FBCNet [29] explicitly integrates a priori information in the frequency and spatial domains, effectively reducing model complexity while improving motor imagery classification performance. This observation highlights that embedding a priori information into the design of deep learning models can achieve a more optimal balance between model complexity and classification accuracy.

Building on the above analysis, this paper proposes a novel hierarchical feature recalibration network (HFRNet) for motor imagery EEG classification. The proposed method fully exploits the hierarchical a priori information inherent in TFS features to construct a hierarchical recalibration framework that effectively addresses the challenges of TFS feature learning while maintaining a balance between model complexity and classification accuracy. First, the intrinsic grouping and hierarchical relationships among TFS features are defined, forming a three-layer tree-like structure. Then, a two-stage weighting network is designed to perform layer-by-layer recalibration of TFS features. Specifically, feature weights are assigned progressively—from time–frequency units to time windows—using squeeze-and-excitation operations [25] to capture interdependencies among grouped features within each level. The effectiveness of the proposed HFRNet is validated through experiments on five publicly available motor imagery EEG datasets, demonstrating its superior performance and generalization capability.

The main contributions and innovations of this paper are summarized below:

1. The hierarchical structure between TFS features is proposed and elaborated in detail for the first time and is employed as a priori information for constructing EEG classification models.
2. A HFRNet is proposed to explicitly model hierarchical structures between TFS features. The proposed hierarchical recalibration framework effectively addresses the challenges of TFS feature learning while maintaining a balance between model complexity and classification accuracy.
3. Extensive experiments are performed on five motor imagery EEG datasets, the proposed method is compared with SENet, 30 feature selection methods, and a large of recent literature. The experimental results show that the proposed method significantly improves the classification performance of motor imagery.

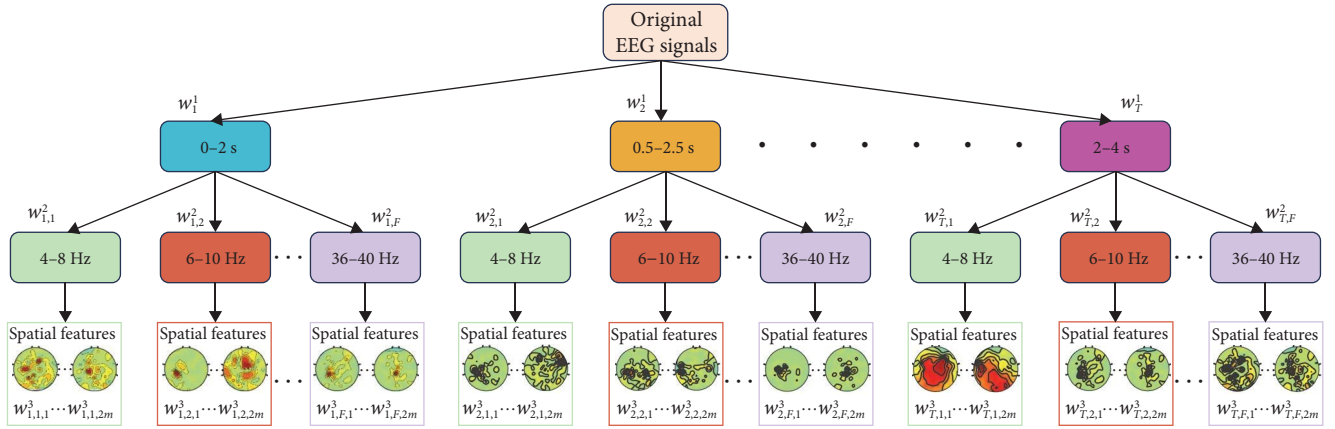


FIGURE 1: The hierarchical structure in TFS features. The feature structure is a three-layered tree distribution, with the first layer being the features for each time window, the second layer being the features for each time–frequency unit, and the third layer being the specific individual spatial features.

The rest of this paper is organized as follows: Section 2 introduces the motor imagery EEG datasets. Section 3 details the hierarchical structure of TFS features and the proposed HFRNet method. Section 4 presents and analyzes the experimental results. Section 5 concludes the paper.

2. The Proposed Methods

We first elaborate on the hierarchical structure of TFS features and then introduce the proposed HFRNet.

2.1. The Hierarchical Structure of TFS Features. The hierarchical structure of TFS features is illustrated in Figure 1. The raw EEG signal is first divided into multiple time windows, after which each time window is further decomposed into several frequency subbands. Spatial features are then extracted from each time–frequency unit. The division of time–frequency units follows the approach described in references [16, 30]. For each time–frequency unit, spatial features are independently extracted using the common spatial pattern (CSP) method. In this study, the number of spatial filter pairs in CSP is set to 1 [31, 32], that is, $m = 1$ in Figure 1. In addition, in Figure 1, $T = 5$ represents the number of time windows and $F = 17$ represents the number of frequency bands.

The entire TFS feature extraction process is performed in a layer-by-layer manner, forming a three-layer tree-structured feature hierarchy. In this structure, the original EEG signal serves as the root node, the time windows and frequency bands act as intermediate nodes (representing the first and second layers of the tree, respectively), and the specific individual spatial features extracted from each time–frequency unit constitute leaf nodes (as the third layers of the tree).

Conventional feature selection methods typically concatenate the spatial features from multiple time–frequency units into a single feature vector. Each feature is then treated as an independent element and assigned an individual weight to capture temporal, frequency, and spatial information. However, this approach overlooks the hierarchical relationships among TFS features. In this study, we leverage this a priori hierarchical information to design a HFRNet that assigns

hierarchical weights to TFS features, thereby preserving and exploiting their intrinsic structural dependencies.

2.2. The Proposed HFRNet. The algorithmic framework of the proposed HFRNet is illustrated in Figure 2. The overall approach can be summarized as follows: first, the features within each time–frequency unit are treated as a single entity, and the first weighting network is constructed to assign weights to the features of each time–frequency unit, which corresponds to $w^2 = [w^2_{1,1}, w^2_{1,2}, \dots, w^2_{1,F}, \dots, w^2_{T,1}, w^2_{T,2}, \dots, w^2_{T,F}]$ in Figure 1. Next, the recalibrated features of each time window are considered as a whole, and the second weighting network is constructed to assign weights to the features of each time window, which corresponds to $w^1 = [w^1_1, w^1_2, \dots, w^1_T]$ in Figure 1.

2.2.1. The First Weighting Network. The extracted TFS features shown in Figure 1, denoted as $X \in R^{H_1 \times W_1 \times C_1}$, are first organized by time–frequency unit. The features of each time–frequency unit are treated as a feature map, where $H_1 = 2m = 2$ and $W_1 = 1$ indicate the dimensions of the feature map, and $C_1 = T \times F = 85$ represents the number of channels across all feature maps.

The squeeze operation is implemented to construct a channel descriptor for the feature maps and can be realized by global average pooling or global maximum pooling [25]. Assuming $z = [z_1, z_2, \dots, z_{C_1}] \in R^{C_1}$ is a channel descriptor, then, the c -th element of z is computed from the corresponding feature map as follows:

$$z_c = \frac{1}{H_1 \times W_1} \sum_{i=1}^{H_1} \sum_{j=1}^{W_1} x_c(i, j), \quad c = 1, 2, \dots, C_1$$

or

$$z_c = \max_{i=1, 2, \dots, H_1, j=1, 2, \dots, W_1} (x_c(i, j)), \quad c = 1, 2, \dots, C_1, \quad (1)$$

where $x_c \in R^{H_1 \times W_1}$ is taken sequentially from $[x_{1,1}, x_{1,2}, \dots, x_{1,F}, \dots, x_{T,1}, x_{T,2}, \dots, x_{T,F}]$.

The excitation operation is used to capture the channel interdependencies among the feature maps, which can be

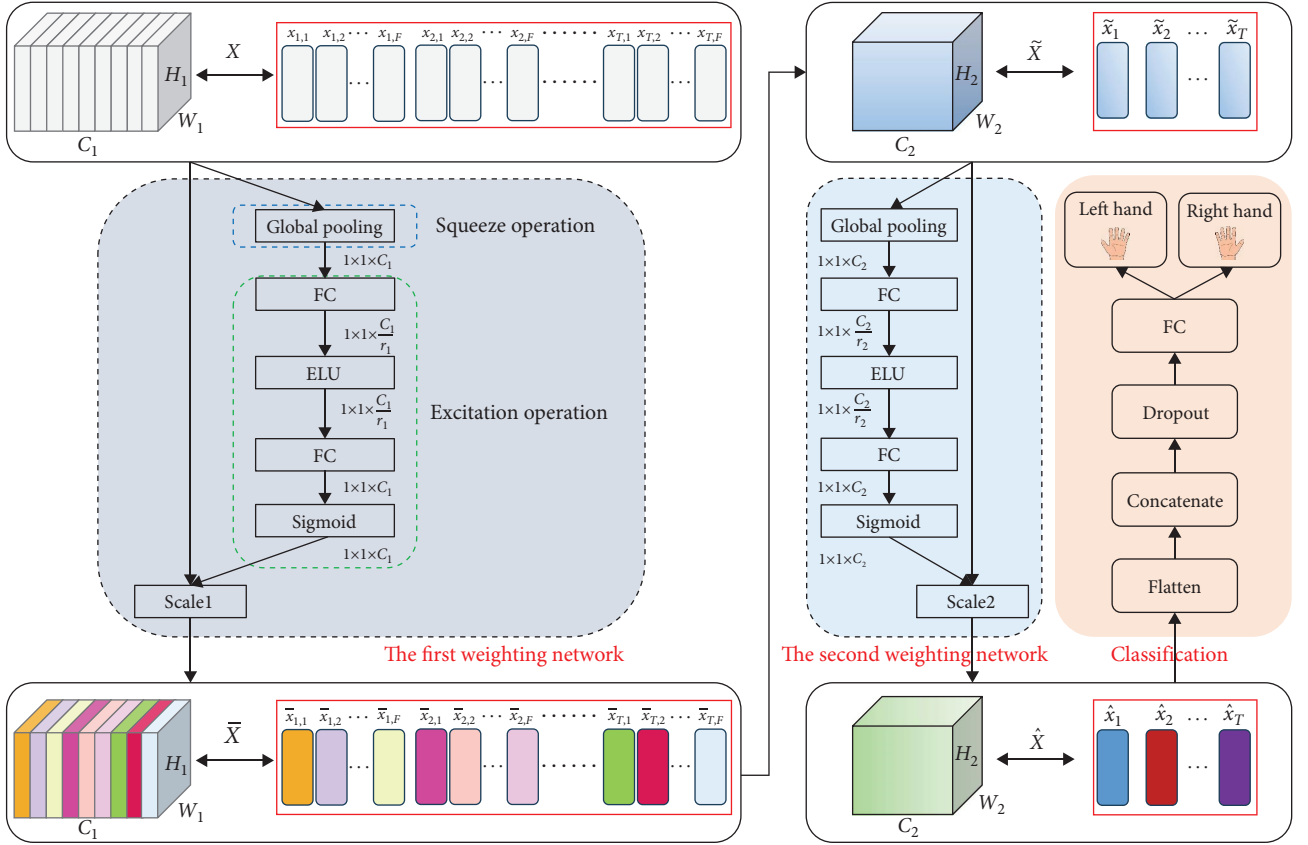


FIGURE 2: The algorithmic framework of the proposed HFRNet. A two-layer weighting network is constructed, one assigns weights to the features of each time–frequency unit and the other assigns weights to the features of each time window.

realized using a bottleneck structure composed of two fully connected (FC) layers [25]. The process is detailed as follows:

$$s = \sigma(W_2 \delta(W_1 z)), \quad (2)$$

where $s \in R^{C_1}$ represents the channel weight, δ refers to the exponential linear unit (ELU) activation function, and σ refers to the sigmoid activation function. $W_1 \in R_{r_1}^{C_1 \times C_1}$ with reduction ratio r_1 is a dimensionality-reduction layer, while $W_2 \in R_{r_1}^{C_1 \times \frac{C_1}{r_1}}$ is a dimensionality-increasing layer. In this paper, r_1 is set to 17.

The updated feature maps are obtained by rescaling x_c with the corresponding channel weight s_c , that is, the original features are recalibrated.

$$\bar{x}_c = F_{\text{scale1}}(x_c, s_c) = s_c \cdot x_c, c = 1, 2, \dots, C_1, \quad (3)$$

where recalibrated features are $\bar{X} = [\bar{x}_1, \bar{x}_2, \dots, \bar{x}_{C_1}]$, F_{scale1} refers to channel-wise multiplication.

2.2.2. The Second Weighting Network. The intermediate recalibrated features $\bar{X}$ outputted by the first weighting network are reorganized by time window, denoted as $\tilde{X} = [\tilde{x}_1, \tilde{x}_2, \dots, \tilde{x}_T] \in R^{H_2 \times W_2 \times C_2}$, $C_2 = T = 5$, $W_2 = 1$, $H_2 = 2m \times F = 34$.

The feature recalibration in the second weighting network follows the same procedure as in the first weighting network. $\tilde{X}$ is sequentially processed through a squeeze operation, an excitation operation, and feature rescaling, except that the reduction ratio is different. In this paper, r_2 is set to 2. The resulting recalibrated feature $\hat{X}$ is then forwarded to the classification block.

2.2.3. Classification Block. In the classification block, feature maps $\hat{X}$ are first flattened and then concatenated into a feature vector, which is subsequently passed through dropout and FC layers. The FC layer contains two neurons with the sigmoid activation function. Binary cross-entropy is used as the loss function, and binary accuracy serves as the evaluation metric.

3. Experimental Results and Analysis

3.1. Experimental Data. Experimental validation is performed on five publicly available motor imagery EEG datasets, sourced from international brain–computer interface competitions and the BNCI Horizon 2020 project. Table 1 provides a summary of the key characteristics of these datasets, with additional details available in the relevant literature or on the official project websites.

For Datasets 1 and 3, which include four types of motor imagery tasks, this study focuses solely on the left-hand and

TABLE 1: Experimental data information.

Datasets	Number of channels	Sampling rate	Number of subjects	Number of tasks	Motor imagery duration	Number of training and test sets for each subject	Data access website
Dataset 1 [33]	60	250 Hz	3 (K3, K6, L1)	4 (left-hand, right-hand, foot, tongue)	4 s	Training set: 90 (K3), 60 (K6, L1); test set: 90 (K3), 60 (K6, L1)	https://www.bbci.de/competition/iii/#data_set_iiia
Dataset 2 [34]	118	100 Hz	5 (aa, al, av, aw, ay)	2 (right-hand, foot)	3.5 s	140, 140 (the training and test sets are recombined)	https://www.bbci.de/competition/iii/#data_set_iva
Dataset 3 [35]	22	250 Hz	9 (A01, A02, ..., A09)	4 (left-hand, right-hand, foot, tongue)	4 s	144, 144	https://www.bbci.de/competition/iv/#dataset2a
Dataset 4 [36]	3	250 Hz	9 (B01, B02, ..., B09)	2 (left-hand, right-hand)	4 s	80, 80 (only the data from the third session)	https://www.bbci.de/competition/iv/#dataset2b
Dataset 5 [37]	15	512 Hz	14 (S01, S02, ..., S14)	2 (right-hand, foot)	4 s	100, 60	http://bnci-horizon-2020.eu/database/data-sets

TABLE 2: Feature selection methods for comparison.

Type	Methods	Abbreviations
Filtered	Fisher criterion	FC [5]
	Mutual information	MI [8]
	Relevant features	Relief [39]
	Pearson correlation coefficient	PCC [6]
	Neighborhood components analysis	NCA [21]
	Minimum-redundancy maximum-relevancy	mRMR [9]
	Variance	VAR [40]
	Chi-square	CS [41]
Wrapped	Binary differential evolution	BDE [42]
	Particle swarm optimization	PSO [43]
	Binary particle swarm optimization	BPSO [43]
	Genetic algorithm	GA [43]
	Binary gray wolf optimization	BGWO [44]
	Atom search optimization	ASO [45]
	Binary dragonfly algorithm	BDA [46]
	Binary tree growth algorithm	BTGA [47]
	Whale optimization algorithm	WOA [48]
	Binary atom search optimization	BASO [49]
	Binary Harris hawk optimization	BHHO [50]
Embedded	Equilibrium optimizer	EO [51]
	Least absolute shrinkage and selection operator	LASSO [16]
	Group LASSO	gLASSO [16]
	Sparse group LASSO	sgLASSO [16]
	Fused LASSO	fLASSO [16]
	Elastic net	EN [16]
	Logarithmic	LOG [19]
	Smoothly clipped absolute deviation	SCAD [52]
	Minimax concave penalty	MCP [53]
	Sparse Bayesian linear discriminant analysis	sBLDA [54]
	Group sparse Bayesian linear discriminant analysis	gsBLDA [54]

right-hand tasks. For Dataset 2, as the small training set issue is not addressed here, the training and test sets are recombined to ensure equal sample sizes. For Dataset 4, only data from the third session are analyzed [15]. For Dataset 5, the original data are downsampled to 256 Hz.

3.2. Model Training. The model is trained using the Adam optimizer with a learning rate of 0.0001 and a dropout rate of 0.5. An early stopping mechanism is employed, halting training if the validation loss does not improve for 20 consecutive epochs. Training runs for a maximum of 1000 epochs with a batch size of 10. The process is executed on an NVIDIA RTX 3090 GPU with 24 GB of memory, using the TensorFlow 2.10 deep learning framework.

For each subject, the original training set was further divided into training and validation sets using the stratified fivefold cross-validation. The model was trained five times to obtain five distinct models, which were then used to classify

and predict the test set. The final classification metrics for each subject were calculated as the average of the results from the five models [38].

3.3. Experimental Results. Classification accuracy and F1 score were used as evaluation metrics. For each subject, these metrics were computed based on the classification results of the test set samples.

3.3.1. HFRNet Compared With Feature Selection Methods. HFRNet is compared with 30 feature selection methods, including eight filtered, 12 wrapped, and 10 embedded methods, as detailed in Table 2.

For filtered methods, features were first ranked according to their importance, and then the top K features were selected for classification. The optimal value K was determined using Fisher linear discriminant analysis (FLDA) [19] combined with tenfold cross-validation.

TABLE 3: HFRNet compared with filtered feature selection methods at the dataset-level.

Feature learning methods	Average classification accuracy					Average F1 score				
	Dataset 1	Dataset 2	Dataset 3	Dataset 4	Dataset 5	Dataset 1	Dataset 2	Dataset 3	Dataset 4	Dataset 5
FC	76.30	83.72	79.86	75.69	71.07	0.7862	0.8106	0.7851	0.7653	0.6915
MI	72.41	79.86	76.78	75.56	73.45	0.6506	0.7724	0.7525	0.7564	0.7048
Relief	77.96	81.57	79.01	73.89	71.55	0.7916	0.7857	0.7704	0.7428	0.7031
PCC	76.30	83.72	79.86	75.69	71.07	0.7862	0.8106	0.7851	0.7653	0.6915
NCA	75.56	81.14	78.78	75.83	77.86	0.7779	0.7801	0.7655	0.7661	0.7689
mRMR	82.22	79.71	79.01	74.17	70.71	0.8358	0.7499	0.7586	0.7373	0.6852
VAR	82.78	81.43	77.08	73.75	72.38	0.8427	0.7995	0.7663	0.7455	0.7258
CS	77.78	62.57	80.71	77.22	74.28	0.7465	0.5212	0.8123	0.7795	0.7254
HFRNet	86.00	83.06	82.36	80.61	78.64	0.7932	0.8168	0.8144	0.8066	0.7622

Note: Bold values indicate the optimal results among the comparative data.

TABLE 4: HFRNet compared with wrapped feature selection methods at the dataset-level.

Feature learning methods	Average classification accuracy					Average F1 score				
	Dataset 1	Dataset 2	Dataset 3	Dataset 4	Dataset 5	Dataset 1	Dataset 2	Dataset 3	Dataset 4	Dataset 5
BDE	68.52	75.29	71.84	64.86	65.6	0.5937	0.6861	0.7361	0.6230	0.6275
PSO	76.85	77.71	71.84	65	64.76	0.7737	0.7224	0.7132	0.6389	0.6389
BPSO	76.85	80.43	72.22	66.39	66.19	0.7737	0.7588	0.7164	0.6650	0.6479
GA	76.85	79.71	70.45	65.14	67.02	0.7737	0.7497	0.7027	0.6389	0.6609
BGWO	76.85	80.14	72.45	67.22	66.43	0.7737	0.7546	0.7145	0.6610	0.6517
ASO	72.96	74.57	71.45	65.28	65.95	0.6799	0.6811	0.7265	0.6445	0.6482
BDA	70.74	76.28	73.22	67.08	64.17	0.6432	0.7125	0.7474	0.6618	0.6188
BTGA	73.33	75.43	73.15	66.67	65.36	0.6971	0.6994	0.7320	0.6555	0.6254
WOA	76.85	80.57	74.77	75	65.24	0.7737	0.7591	0.7253	0.7523	0.6427
BASO	76.85	80.14	72.69	67.36	67.38	0.7737	0.7546	0.7103	0.6537	0.6620
BHHO	68.89	77	72.76	66.67	64.17	0.6258	0.7297	0.7337	0.6562	0.6197
EO	76.85	80.43	72.76	66.25	67.14	0.7737	0.7574	0.7307	0.6622	0.6570
HFRNet	86.00	83.06	82.36	80.61	78.64	0.7932	0.8168	0.8144	0.8066	0.7622

Note: Bold values indicate the optimal results among the comparative data.

For wrapped methods, parameters were set according to the respective references. Once the optimal feature subset was identified, classification was performed using the FLDA classifier.

For embedded methods, feature selection and classification are performed simultaneously, eliminating the need for additional classifiers. The optimal regularization parameters were determined via grid search combined with tenfold cross-validation, the alternative sets of regularization parameters were $\{2^{-5}, 2^{-4.8}, \dots, 2^{4.8}, 2^5\}$. The model parameter a for LOG was set to 0.02, and the model parameter r for SCAD and MCP was set to 3 and 2, respectively.

We compare the classification performance of all methods at both the dataset and subject levels. Tables 3–5 show the dataset-level results. Due to space limitations, only the average classification accuracy for each dataset is reported, calculated as the mean accuracy across all subjects within the dataset. The best results are shown in bold.

As shown in Table 3, compared with filtered methods, HFRNet achieved the highest average classification accuracy on four datasets and the highest average F1 score on three datasets. In comparison with wrapped methods (Table 4),

HFRNet attained the highest average classification accuracy and F1 score across all datasets. Relative to embedded methods (Table 5), HFRNet achieved the highest average classification accuracy and F1 score on two datasets, namely, Dataset 2 and Dataset 5; for the remaining datasets, its performance was slightly below the best but remained competitive. Overall, these results demonstrate that HFRNet outperforms traditional feature selection methods.

The overall average accuracy results and statistical test across all datasets is shown in Table 6, HFRNet achieved the highest average classification accuracy of 81.03% and the highest average F1 score of 0.793. Furthermore, paired t -tests were conducted to compare HFRNet with other methods. Except for the LOG method, HFRNet demonstrated significantly better performance than all other feature selection methods, with p -values smaller than 0.05.

The classification results at the subject-level are shown in Figures 3–5. As shown in Figures 3–5, orange indicates classification accuracy and blue indicates the F1 score, while different marker shapes correspond to subjects from different datasets. Each coordinate point represents the classification accuracy of an individual subject. The vertical axis denotes

TABLE 5: HFRNet compared with embedded feature selection methods at the dataset-level.

Feature learning methods	Average classification accuracy					Average F1 score				
	Dataset 1	Dataset 2	Dataset 3	Dataset 4	Dataset 5	Dataset 1	Dataset 2	Dataset 3	Dataset 4	Dataset 5
LASSO	84.63	78.14	78.86	75.97	72.98	0.8286	0.7192	0.7716	0.7661	0.7119
gLASSO	82.96	77.57	81.17	75	72.38	0.7927	0.7118	0.7857	0.7523	0.7092
sgLASSO	84.07	78.57	79.4	75.97	73.45	0.8194	0.728	0.7722	0.7657	0.7188
flASSO	82.41	78.71	78.78	76.81	73.21	0.8015	0.7271	0.7634	0.7767	0.7151
ElasticNet	83.89	77.57	78.93	77.64	73.09	0.8208	0.7121	0.7642	0.7794	0.7151
LOG	82.96	77.57	83.18	80.14	77.86	0.8111	0.7039	0.8207	0.8024	0.7464
SCAD	86.3	74.86	80.79	80.69	76.07	0.8123	0.6687	0.7945	0.8077	0.7334
MCP	86.3	73.86	81.25	80	75.6	0.8123	0.6494	0.7981	0.8012	0.7258
sBLDA	74.63	79.57	74.46	68.89	68.93	0.7071	0.756	0.735	0.6848	0.6605
gsBLDA	70.37	79.29	75.69	71.94	70.24	0.6305	0.7441	0.7611	0.7233	0.6798
HFRNet	86.00	83.06	82.36	80.61	78.64	0.7932	0.8168	0.8144	0.8066	0.7622

Note: Bold values indicate the optimal results among the comparative data.

TABLE 6: Overall average classification results and statistical test across all datasets.

Methods	Accuracy	F1 score	<i>p</i> -Value	Methods	Accuracy	F1 score	<i>p</i> -Value	Methods	Accuracy	F1 score	<i>p</i> -Value
FC	76.06	0.7511	0.0015	BPSO	70.17	0.6905	1.63E-08	LASSO	76.49	0.7472	3.92E-06
MI	75.40	0.7315	0.0002	GA	69.69	0.6849	9.19E-09	gLASSO	76.39	0.7427	3.54E-05
Relief	75.49	0.7441	1.22E-05	BGWO	70.46	0.6899	1.55E-08	sgLASSO	76.79	0.7501	1.36E-05
PCC	76.06	0.7511	0.0015	ASO	68.64	0.6715	2.54E-11	flASSO	76.65	0.7478	1.87E-05
NCA	77.85	0.7696	0.0026	BDA	68.87	0.6710	1.82E-09	ElasticNet	76.80	0.7482	4.66E-05
mRMR	75.35	0.7328	2.20E-05	BTGA	69.26	0.6708	6.50E-09	LOG	79.91	0.7753	0.2766
VAR	75.66	0.7573	0.0002	WOA	72.37	0.7103	2.03E-06	SCAD	78.79	0.7617	0.0273
CS	75.19	0.7332	0.0005	BASO	70.88	0.6909	1.04E-08	MCP	78.44	0.7560	0.0212
BDE	68.26	0.6557	6.09E-10	BHHO	68.62	0.6678	9.39E-11	sBLDA	71.92	0.6982	5.73E-08
PSO	68.93	0.6762	2.26E-09	EO	70.60	0.6961	1.89E-08	gsBLDA	72.99	0.7122	1.43E-07
HFRNet	81.03	0.7931	—	HFRNet	81.03	0.7931	—	HFRNet	81.03	0.7931	—

Note: Bold values indicate the optimal results among the comparative data.

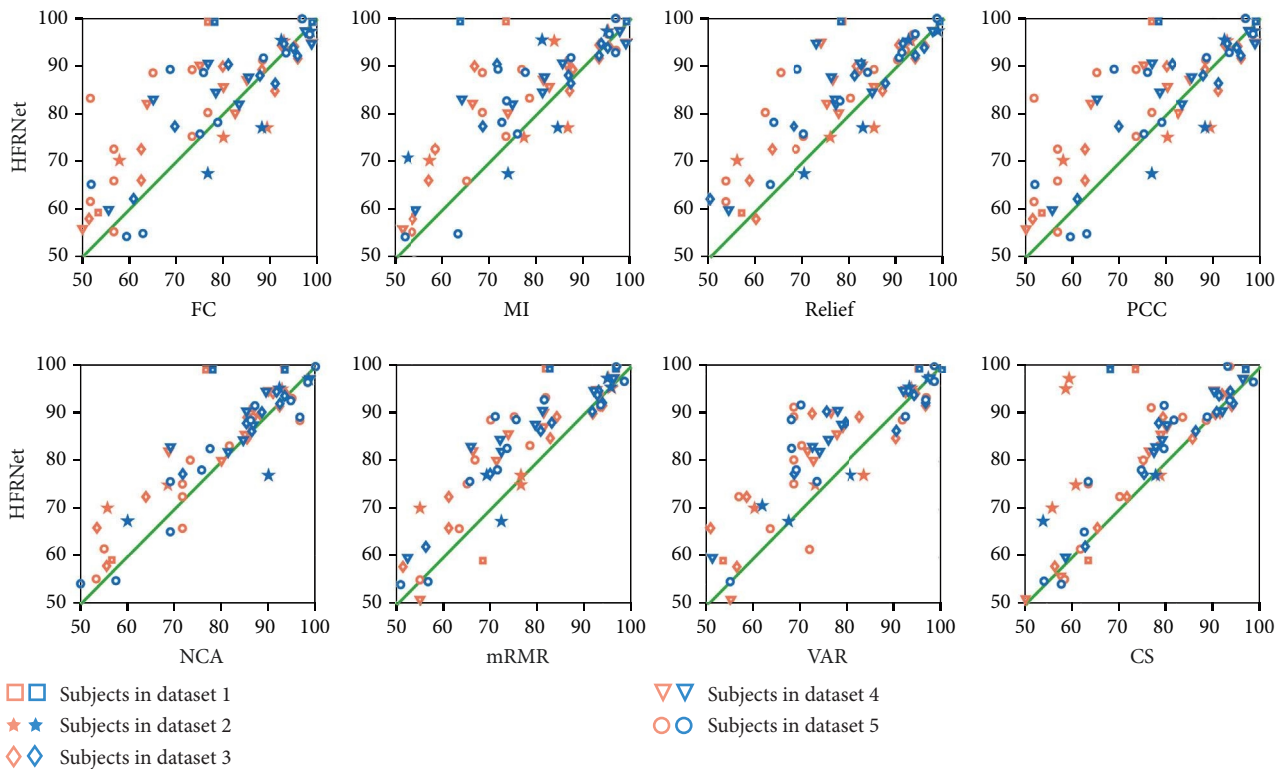


FIGURE 3: HFRNet compared with filtered feature selection methods at the subject-level. Each plotted point corresponds to the classification accuracies of HFRNet and the comparison methods for one subject. The vertical axis denotes the classification accuracy of HFRNet, while the horizontal axis corresponds to that of the comparison methods.

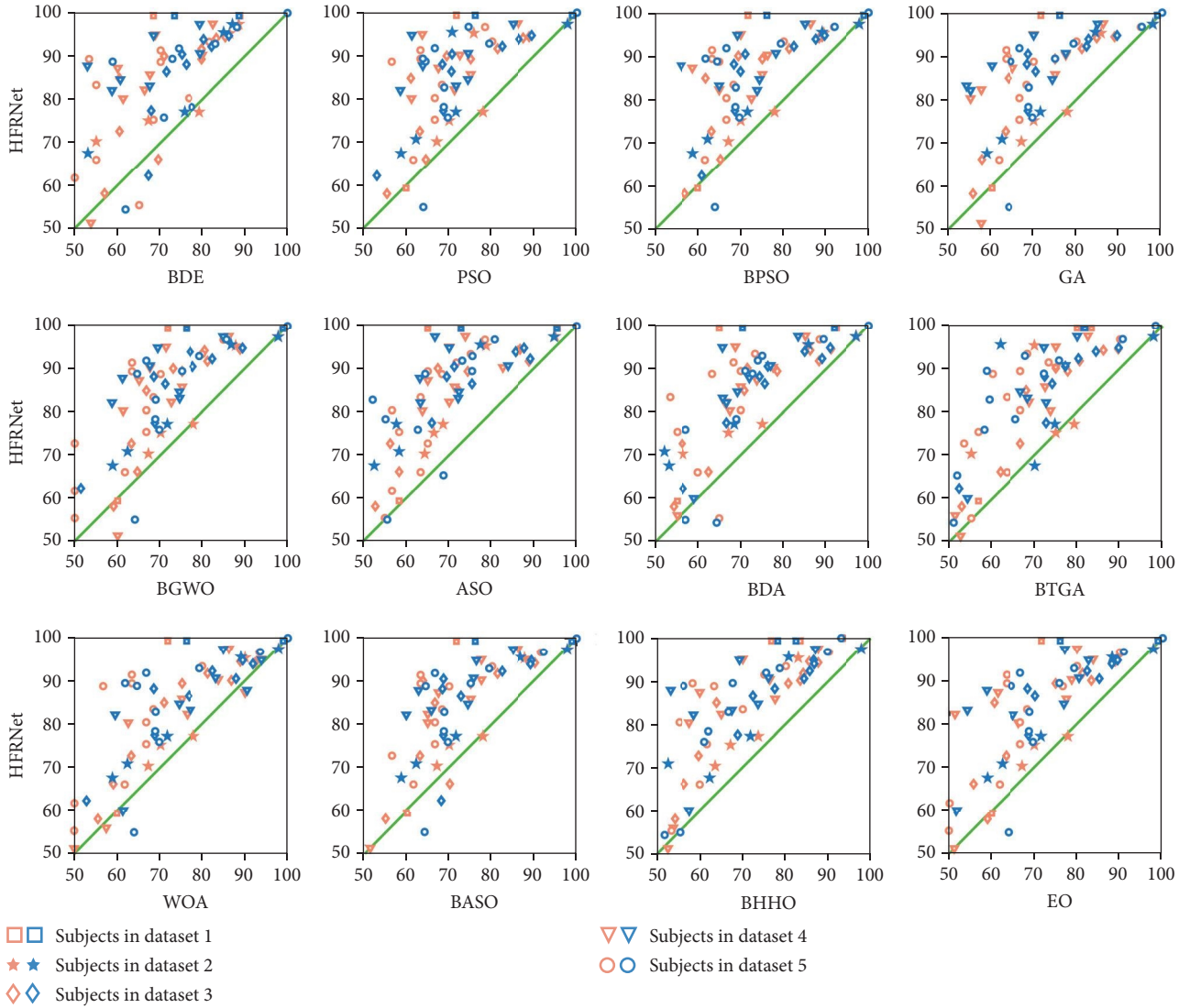


FIGURE 4: HFRNet compared with wrapped feature selection methods at the subject-level. Each plotted point corresponds to the classification accuracies of HFRNet and the comparison methods for one subject.

the classification accuracy of HFRNet, while the horizontal axis corresponds to that of the comparison methods. Points located along the diagonal represent identical classification results between HFRNet and the comparison method, whereas points above and to the left of the diagonal indicate superior performance by HFRNet. As shown in Figures 3–5, HFRNet outperforms the comparison methods for the majority of subjects, further demonstrating the effectiveness of the proposed model.

3.3.2. HFRNet Compared With the Recent Literature. To more comprehensively assess the classification performance of the proposed method, we compared it with recent approaches from the literature, including traditional methods (the first three columns of the comparison methods) and network-based methods (the last three columns of the comparison methods). Traditional methods perform feature extraction and selection to construct feature samples, which are

subsequently classified using machine learning algorithms. In contrast, network-based methods rely mainly on deep learning architectures—such as transformer- or graph-based EEG models—that integrate feature extraction, selection, and classification into a unified framework. As shown in Table 7, our method outperforms most existing approaches, exhibiting particularly strong advantages over network-based methods.

3.3.3. HFRNet Compared With SENet. HFRNet is compared with the SENet [25], and the results are shown in Figure 6. In this paper, SENet consists of the first weighting network and the classification block in Figure 2, that is, the intermediate feature $\bar{X}$ is directly fed to the classification block.

As shown in Figure 6, HFRNet improves the classification performance of SENet, particularly for Dataset 1. These results highlight the benefit of learning hierarchical structures among TFS features.

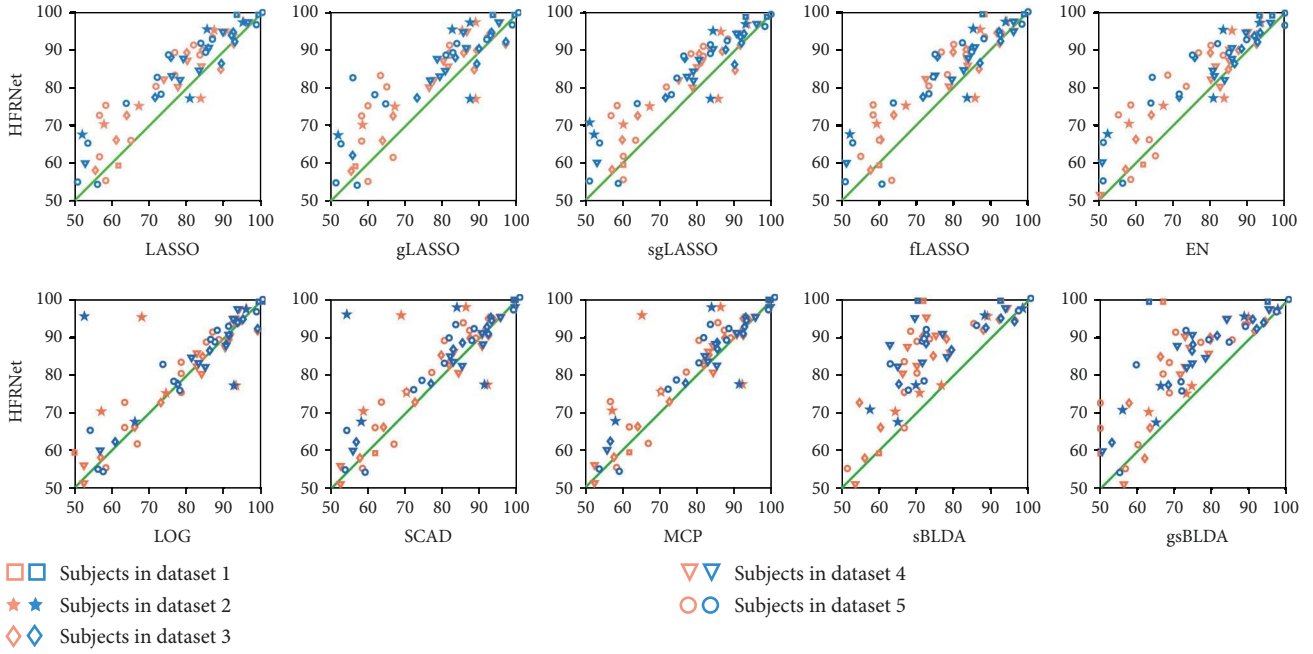


FIGURE 5: HFRNet compared with embedded feature selection methods at the subject-level. Each plotted point corresponds to the classification accuracies of HFRNet and the comparison methods for one subject.

4. Discussion

The proposed method requires setting only two network parameters: r_1 and r_2 . The parameter r_1 is the reduction ratio for the first weight network, which we set to 17, primarily based on the original SENet paper, where a reduction ratio of 16 was found to be optimal. In our case, however, the number of neurons in the previous layer is 85 (corresponding to the number of time–frequency units), which is not divisible by 16. To ensure divisibility and achieve better experimental outcomes, we set r_1 to 17. The parameter r_2 is the reduction ratio for the second weight network, where the previous layer has only five neurons (corresponding to the number of time windows). Setting r_2 to 1 would have no compression effect, while setting it to 3 would overly compress the next layer, potentially reducing the number of neurons to just 1 neuron. To avoid excessive compression, we set r_2 to 2.

In our experiments, we found that both TFS feature normalization and the squeeze operation affect the performance of HFRNet. First, we examined the impact of TFS feature normalization on the classification results of HFRNet using GMP squeeze operation, with the results shown in Figure 7. Feature normalization was performed using the svm-scale function in the LIBSVM toolkit [81], which scales the features in each dimension to the range 0–1. As illustrated in Figure 7, feature normalization generally reduces the classification performance of HFRNet, except for Dataset 1.

Based on our analysis, we concluded that feature normalization affects the squeeze operation in HFRNet. While feature normalization is applied to each one-dimensional feature individually, the squeeze operation considers all feature dimensions

simultaneously. As a result, normalization can alter the relative scaling between adjacent feature dimensions, leading to different channel descriptors during the squeeze operation.

Without feature normalization, we examined the impact of different squeeze operations on the classification results of HFRNet. Specifically, GAP and GMP squeeze operations were compared, with the results shown in Figure 8. As illustrated in Figure 8, GMP outperforms GAP for all datasets except Dataset 2.

Based on the working principle of CSP, it is known that CSP maximizes the variance of one type of task while minimizing the variance of the other [31, 32]. Consequently, the maximization operation more effectively select the channel features with larger variance, whereas the averaging operation tends to smooth the features.

Based on the above experimental results, we conclude that HFRNet achieves superior classification performance when feature normalization is not applied and the GMP squeeze operation is used.

5. Conclusion

A HFRNet is proposed to model the grouping and hierarchical structure of TFS features for motor imagery classification. First, the grouping and hierarchical structure of TFS features are elaborated, followed by the construction of a two-layer weighting network that recalibrates the TFS features layer by layer. The experimental results show that the proposed method outperforms the existing feature recalibration network (SENet), surpasses traditional feature selection methods, and exceeds the performance of most recent approaches

TABLE 7: Classification accuracy of the proposed method and the recent literature.

Dataset	Compared methods (year)					HfRENet	
Dataset 1	DASS [55] (2019) 85.00	Djelloul and Belkacem [56] (2021) 76.40	Zhou et al. [57] (2023) 77.30	SF-2DCNN [58] (2021) 77.78	ANN [59] (2022) 68.90	MNCS [60] (2024) 79.40	86.00
Dataset 2	R-AdaB [61] (2019) 80.60	SVM-CCA-CS [62] (2020) 80.03	ACSP [63] (2022) 82.7	CNN-SAE [64] (2020) 82.00	Comformer [65] (2022) 81.71	MSVTNet [66] (2024) 76.46	83.06
Dataset 3	R-MDRM [67] (2019) 80.98	Guan et al. [68] (2019) 79.41	Cecotti [69] (2023) 74.73	Graph-CSPNet [1] (2023) 78.82	MCSNet [70] (2023) 74.10	STFEnc [71] (2025) 79.51	82.36
Dataset 4	STFTCOV [4] (2022) 78.69	FBRK-SVM [72] (2022) 76.21	RCM [73] (2023) 72.01	CNN-VAE [74] (2021) 78.20	DFormer [75] (2024) 76.18	TBTSCtNet [76] (2024) 78.20	80.61
Dataset 5	Wahid and Tafreshi [77] (2021) 78.90	RCM [73] (2023) 70.52	Carrara Papadopoulou [78] (2024) 70.00	Snrkovsky and Cecotti [79] (2023) 73.75	Shallow ConvNet [78] (2024) 64.00	MVCNet [80] (2025) 76.37	78.64

Note: Bold values indicate the optimal results among the comparative data.

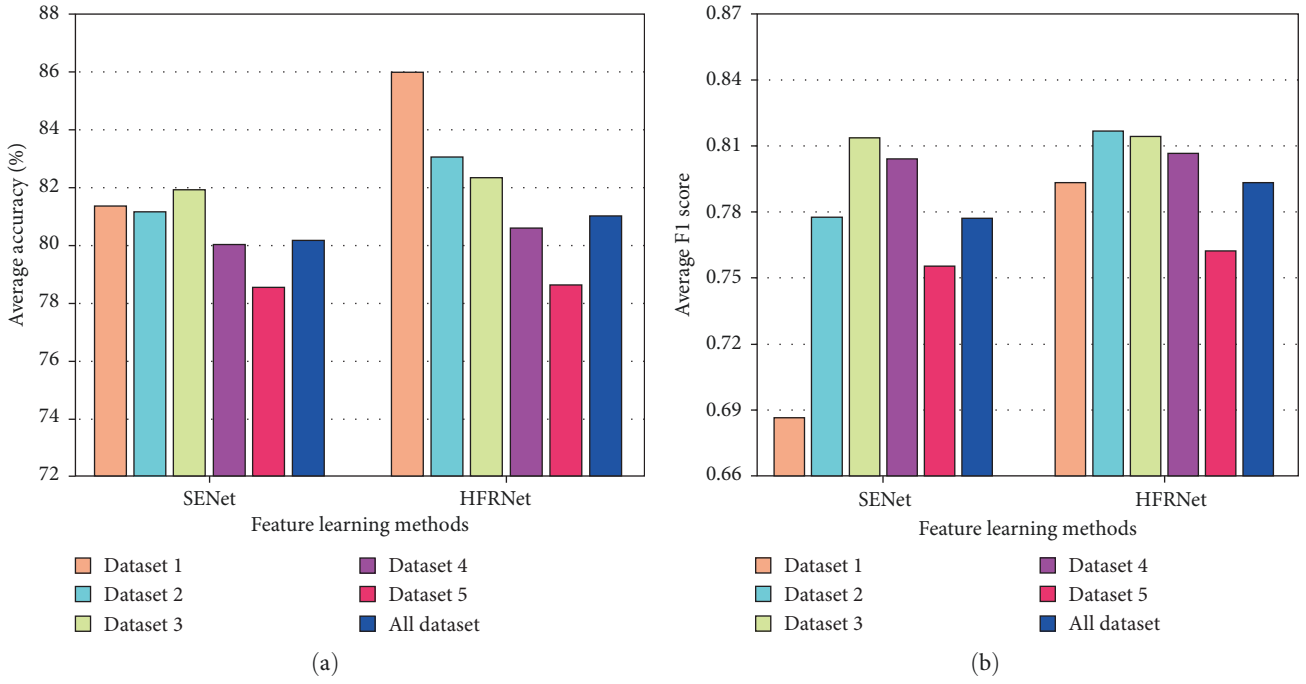


FIGURE 6: The classification results of SENet and HFRNet across all datasets. (a) Classification accuracy and (b) F1 score.

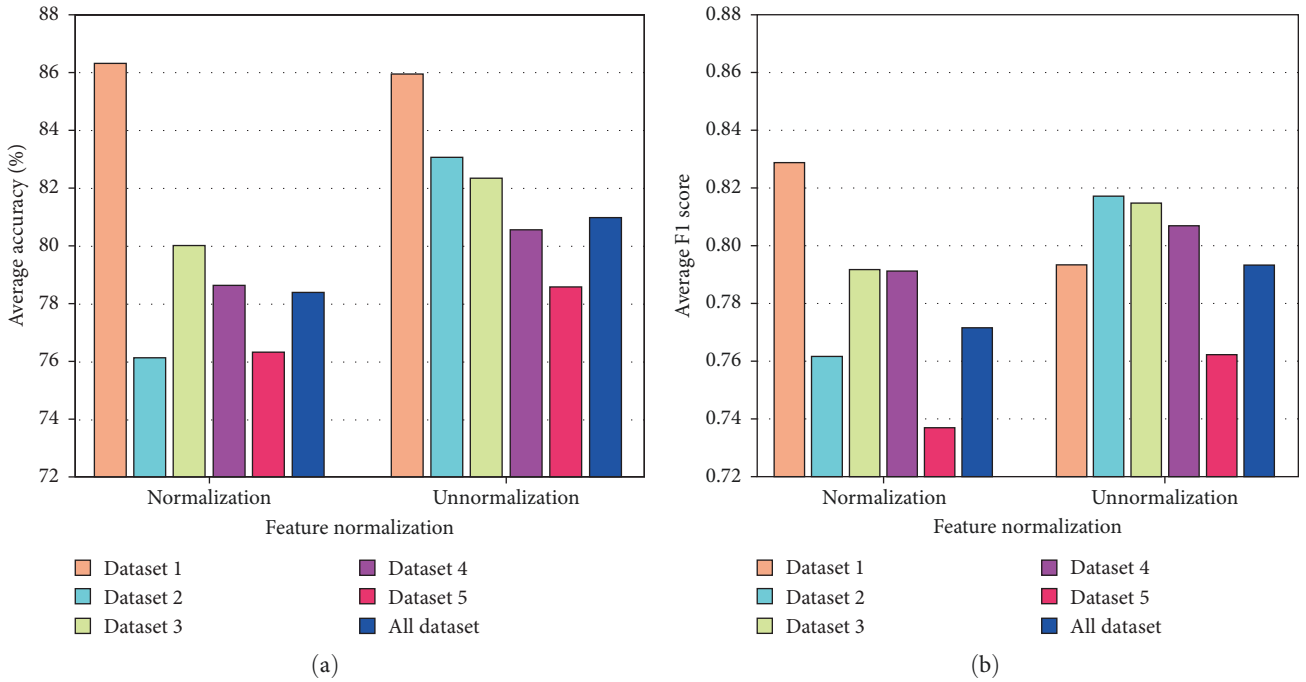


FIGURE 7: The impact of feature normalization on the classification results of HFRNet. (a) Classification accuracy and (b) F1 score.

reported in the literature. The code for HFRNet will be released on GitHub.

In this study, we focused solely on modeling feature relationships between channels, without considering intrachannel relationships within the weighting network. In future work, we

plan to design new networks that more accurately capture the prior information in TFS features to further enhance motor imagery EEG classification performance. Furthermore, the performance of the proposed method under cross-subject evaluation has not yet been evaluated.

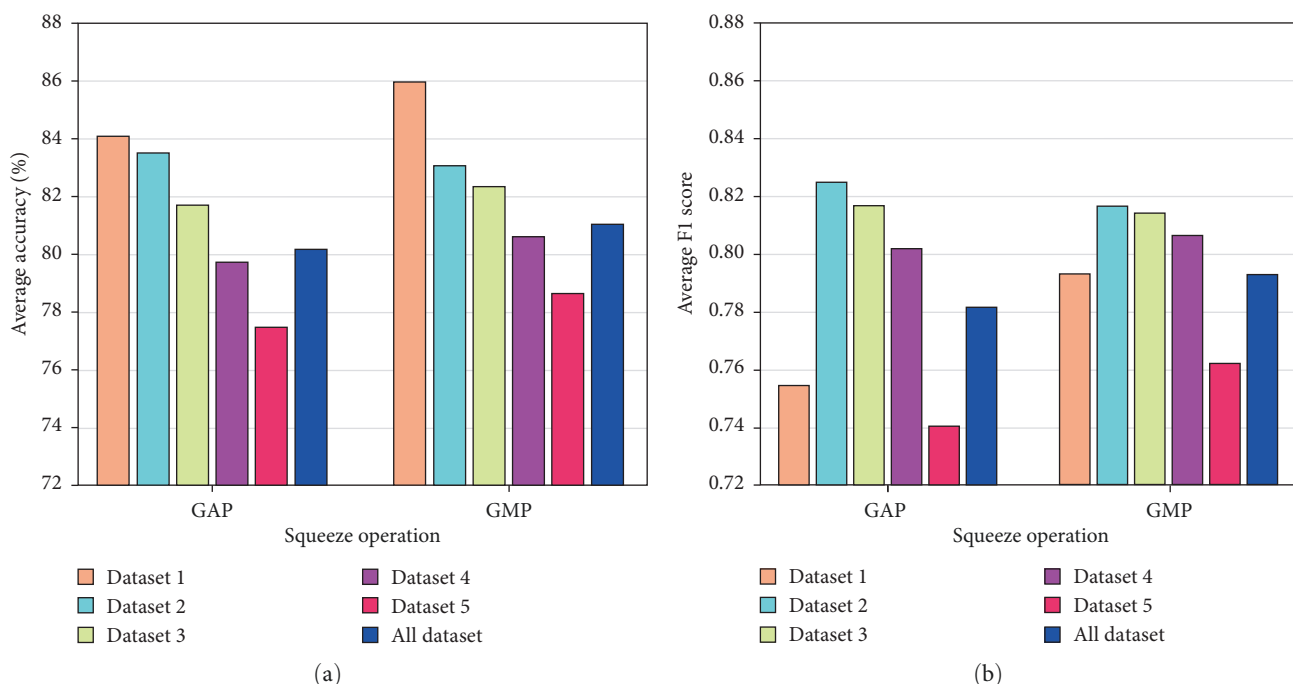


FIGURE 8: The impact of squeeze operation on the classification results of HFRNet. (a) Classification accuracy and (b) F1 score.

Data Availability Statement

The publicly available datasets are used. The data description section in this paper provides links to data access.

Ethics Statement

The authors have nothing to report.

Disclosure

All authors read and approved the final manuscript.

Conflicts of Interest

The authors declare no conflicts of interest.

Author Contributions

Shaorong Zhang and Gan Huang contributed to the study concept and design. Yi Li and Benxin Zhang performed algorithm optimization. Zhen Liang, Li Zhang, LinLing Li, Zhiguo Zhang, Bao Feng, and Tianyou Yu conducted the collection, analysis, and interpretation of data. Shaorong Zhang and Gan Huang worked on the preparation of the manuscript.

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